BMW

Global Sales Analytics

Exploratory Data Analysis Report

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Dataset Source

Link to Tableau Dashboard

1. Introduction

This report presents an exploratory data analysis of BMW Group's global sales performance between 2018 and 2025. The dataset covers four geographic regions — China, Europe, USA, and Rest of World — and eight vehicle models including both traditional combustion-engine vehicles and BMW's growing battery electric vehicle (BEV) lineup. All visualizations were built using Tableau Public.

1.1 Research Questions

Four research questions were identified at the start of the analysis, each targeting a different aspect of the dataset:

- Q1: How have BMW's global unit sales changed over time, and which region grew the fastest?
- Q2: Does BEV adoption vary by region, and which market is leading BMW's electric vehicle transition?
- Q3: How does macroeconomic GDP growth correlate with BMW's annual sales volumes?
- Q4: Which months and regions consistently generate the highest revenue, and is there a seasonal pattern?

2. Dashboard Overview

The following dashboard brings together some of the visualizations in a single view, with KPI cards at the top providing headline metrics and interactive filters allowing the user to explore data by region.

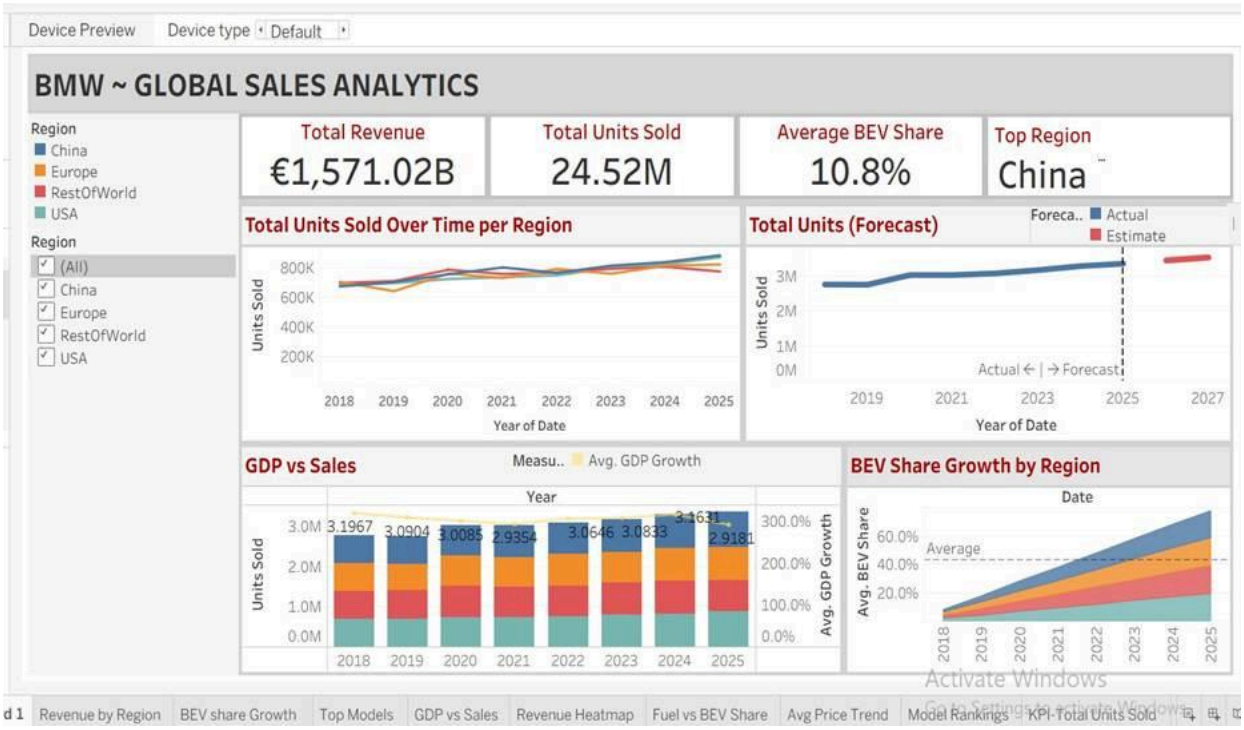

Figure 0: *BMW Group Global Sales Analytics — Interactive Dashboard Overview (2018–2025)*

The four KPI cards confirm the scale of BMW's operations: 24.52 million total units sold, an estimated €1,571 billion in cumulative revenue, an average BEV share of 10.8% across the full period, and China identified as the top-performing region by volume.

Total Units Sold	Total Revenue	Avg BEV Share	Top Region
24.52M	**€1,571B**	**10.8%**	**China**
2018–2025 cumulative	Estimated aggregate	Rising to ~20% by 2025	Highest volume market

3. Data Quality Assessment

Before building any visualizations, the dataset was examined using to check its quality and identify any issues that could affect the analysis. The checks below were conducted using Python (pandas library) and Tableau's built-in data preview tools.

3.1 Dataset Overview

- Total rows: 3,072 records (4 regions × 8 models × 96 months)
- Total columns: 11 fields covering time, geography, volume, pricing, and macro indicators
- Date range: January 2018 to December 2025 (8 full years)
- No missing values were found in any field
- No duplicate rows were detected

4. Visualizations

4.1 Total Units Sold Over Time per Region

How have BMW's global unit sales changed over time, and which region grew the fastest?

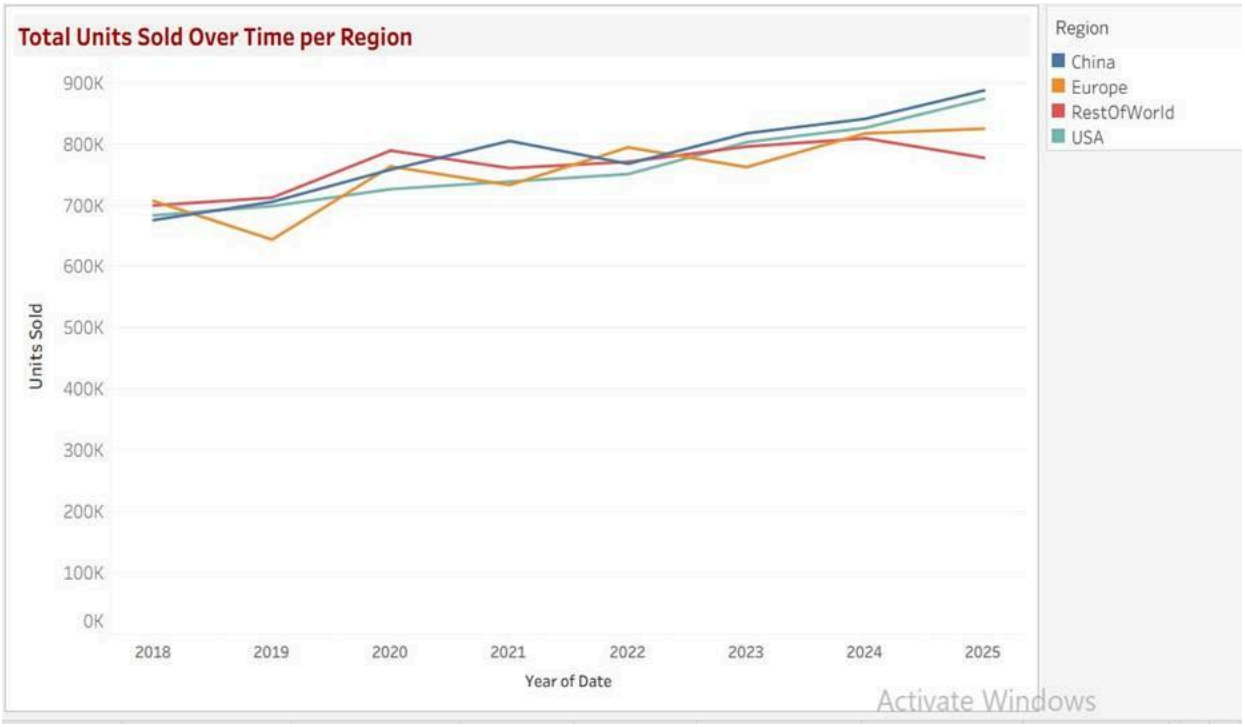

Figure 1: *Total Units Sold Over Time per Region — Line Chart (2018–2025)*

This line chart shows the annual sales trajectory for each of BMW's four regions from 2018 to 2025. All four regions display a positive overall growth trend, confirming that BMW expanded its global sales volume consistently throughout the period. China (blue) emerges as the dominant and fastest-growing region, overtaking Europe and Rest of World to reach nearly 900K annual units by 2025. The chart also reveals a visible dip across most regions in 2020, which directly corresponds to the global disruption caused by the COVID-19 pandemic, before a strong recovery in 2021 and continued growth thereafter.

4.2 Revenue by Region

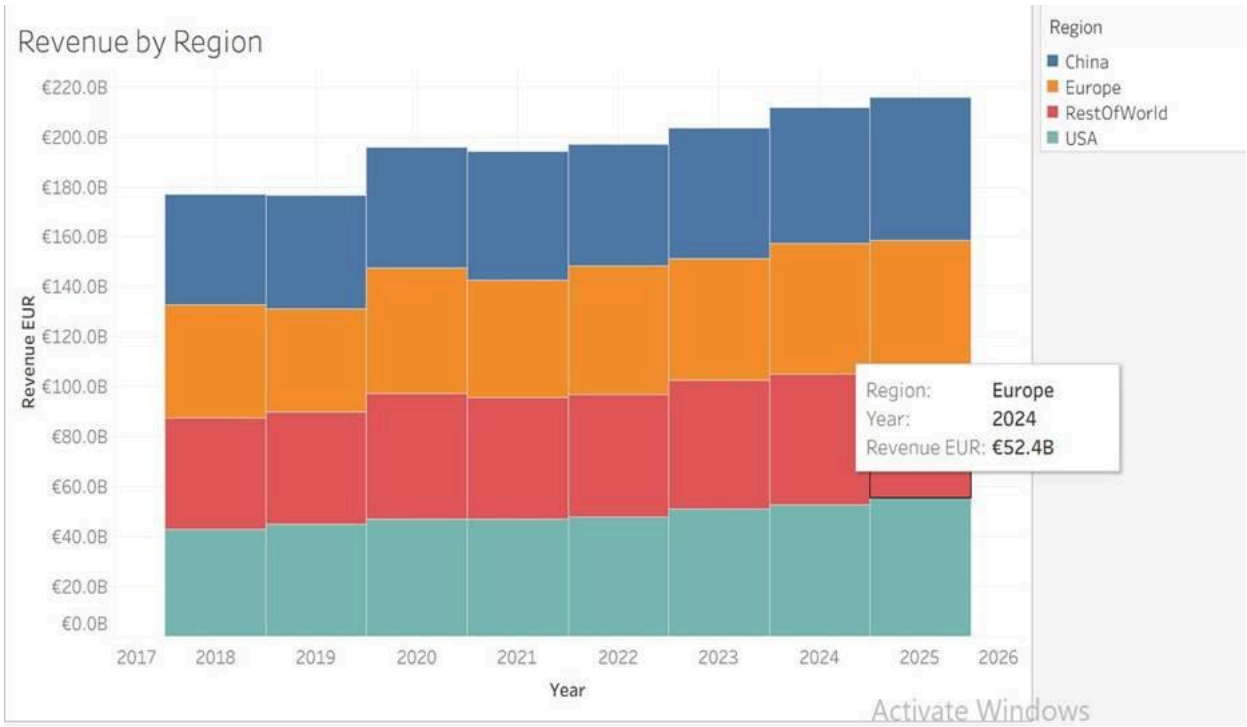

Figure 2: *Revenue by Region — Stacked Bar Chart (2018–2025)*

The stacked bar chart illustrates how BMW's estimated annual revenue grew across all four regions from 2018 to 2025. Total revenue increased from approximately €180B in 2018 to over €215B by 2025, reflecting both unit volume growth and rising average transaction prices. China's blue segment visibly expands each year, becoming the single largest revenue contributor by 2023. Europe (orange) shows steady growth while USA (teal) and Rest of World (red) maintain relatively stable proportions, suggesting that China's growth is the primary driver of BMW's overall revenue expansion during this period.

4.3 BEV Share Growth by Region

Does BEV adoption vary by region, and which market is leading BMW's electric vehicle transition?

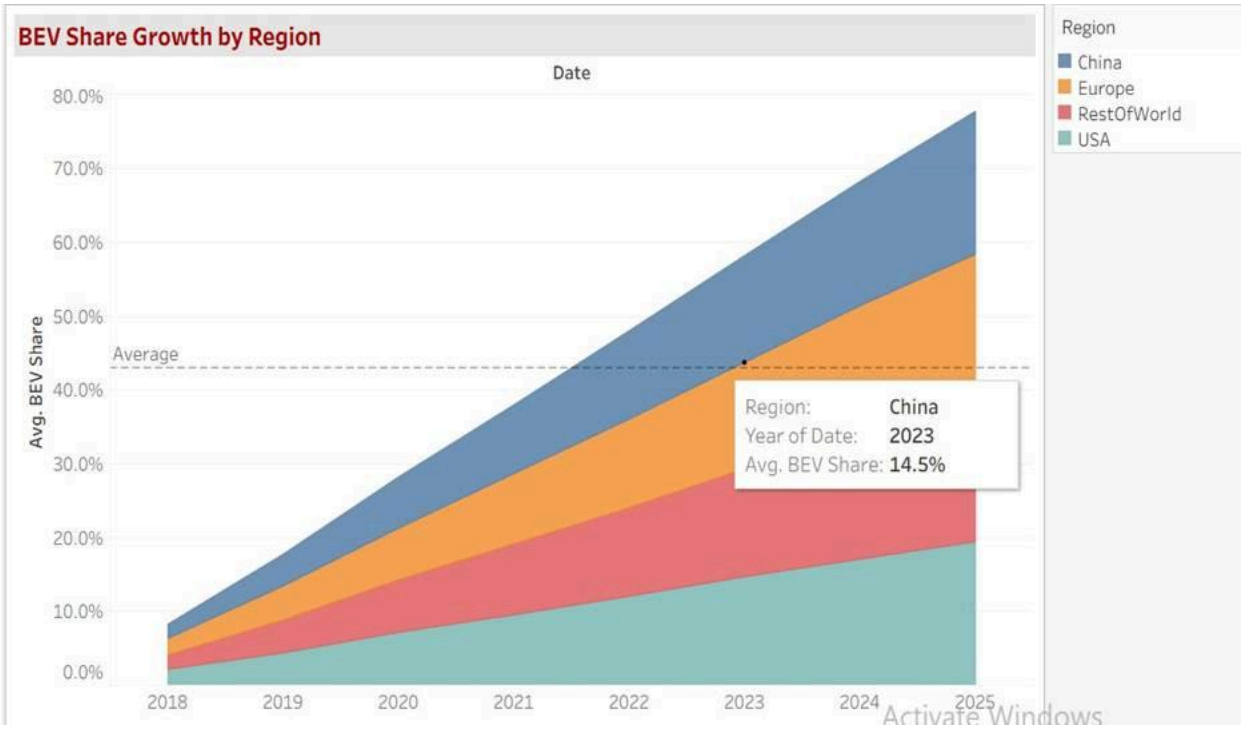

Figure 3: *BEV Share Growth by Region — Stacked Area Chart (2018–2025)*

This stacked area chart tracks the growth in battery electric vehicle share across all four regions from 2018 to 2025. BEV adoption started from near-zero in 2018 and grew exponentially, with the combined average surpassing 40% by 2025 as indicated by the dashed reference line. China leads all regions in BEV adoption, reaching approximately 14.5% in 2023 alone, driven by strong government EV policy incentives and infrastructure investment. This chart directly answers the second research question, confirming that BEV adoption varies significantly by region and that China is the clear leader in BMW's electric vehicle transition.

4.4 Top Models by Units Sold

Figure 4: *Top Models by Units Sold — Horizontal Stacked Bar Chart (2018–2025)*

The stacked bar chart ranks BMW's eight models by total cumulative units sold across all regions, with bars split by region to show geographic contribution. The iX leads in total units, closely followed by i4 and X7, suggesting that BMW's newer models have achieved strong global uptake. Notably, all models show a broadly even regional split, indicating that BMW's portfolio has relatively balanced global appeal rather than being concentrated in a single market.

4.5 GDP Growth vs Units Sold

How does macroeconomic GDP growth correlate with BMW's annual sales volumes?

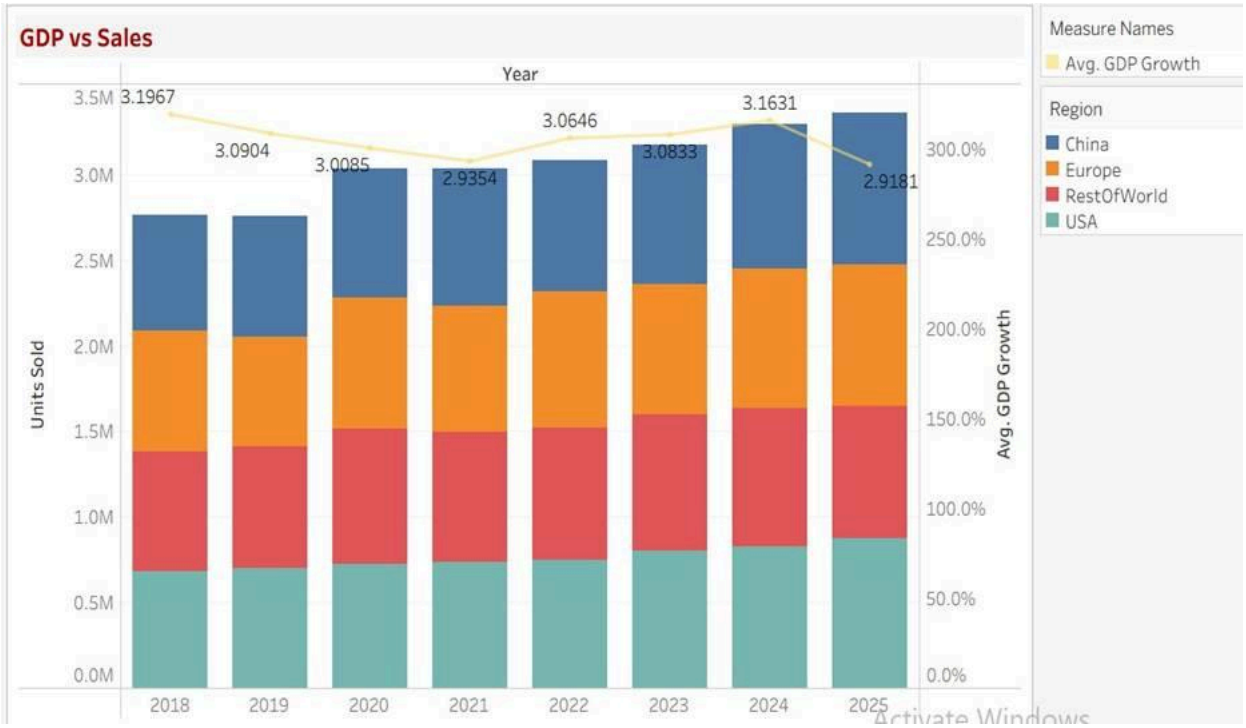

Figure 5: *GDP Growth vs Units Sold — Dual-Axis Chart (2018–2025)*

This chart shows BMW's total units sold each year, with a gold line showing the average GDP growth rate on top. GDP growth stayed fairly steady between 2018 and 2024, sitting around 3.0% to 3.2%. What is interesting is that in 2025, the GDP line drops to its lowest point of around 2.9%, yet BMW actually recorded its highest ever units sold in the same year. This suggests that BMW's sales do not depend heavily on economic growth even when the economy slows down, people still buy BMW cars.

Based on the above discoveries, the follow up question would be :

If BMW's sales are not heavily affected by mild GDP slowdowns, what economic conditions would actually cause a significant drop in BMW sales?

BMW's sales appear resilient to mild economic slowdowns, this is evident because sales hit their highest point despite GDP growth being at its lowest in 2025. The only real drop in sales was in 2020, caused by COVID-19 lockdowns rather than economic conditions alone. This suggests that physical disruptions pose a greater risk to BMW sales than a weakening economy.

4.6 Revenue Heatmap

Which months and regions consistently generate the highest revenue, and is there a seasonal pattern?

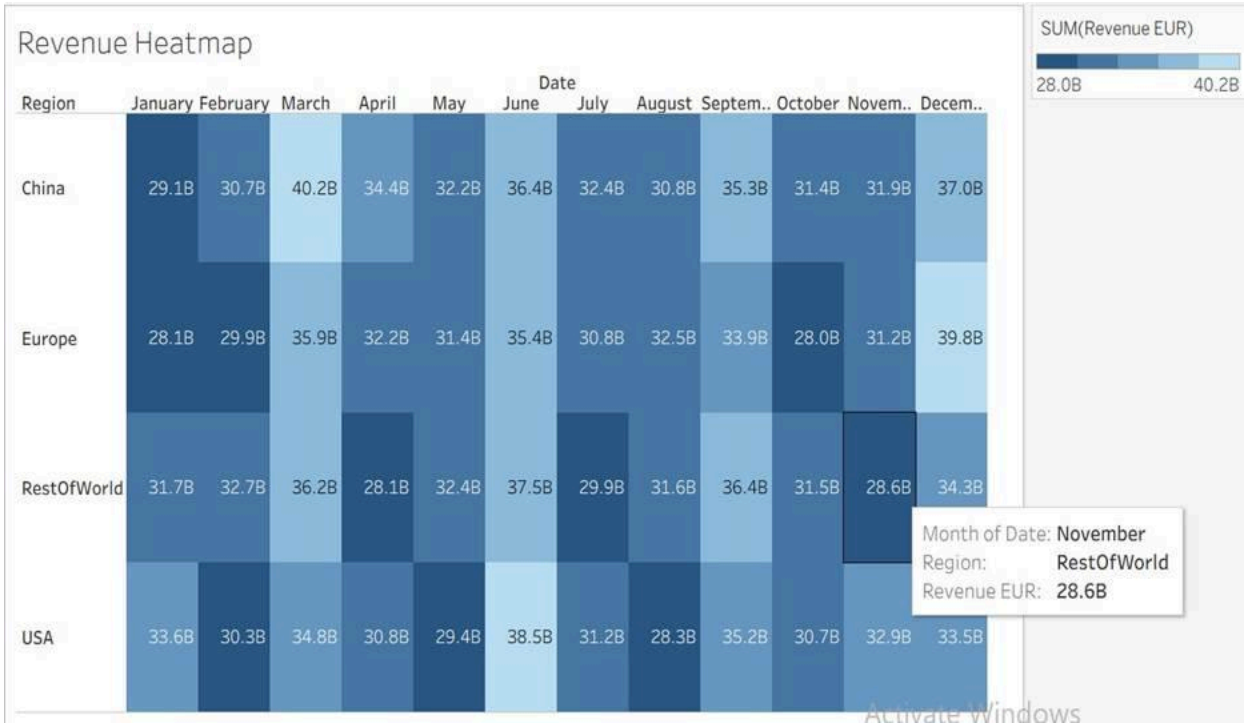

Figure 6: *Revenue Heatmap by Region and Month (2018–2025 Aggregate)*

The revenue heatmap uses colour intensity to show aggregated monthly revenue across all four regions, with lighter blue cells indicating higher revenue months. March stands out as a consistently high-revenue month across all regions, December also shows strong performance, particularly for Europe (€39.8B) and China (€37.0B), confirming year-end purchasing surges. The USA row shows relatively even distribution across months, suggesting a more stable year-round buying pattern compared to the more seasonal European and Chinese markets.

4.7 Fuel Price vs BEV Share

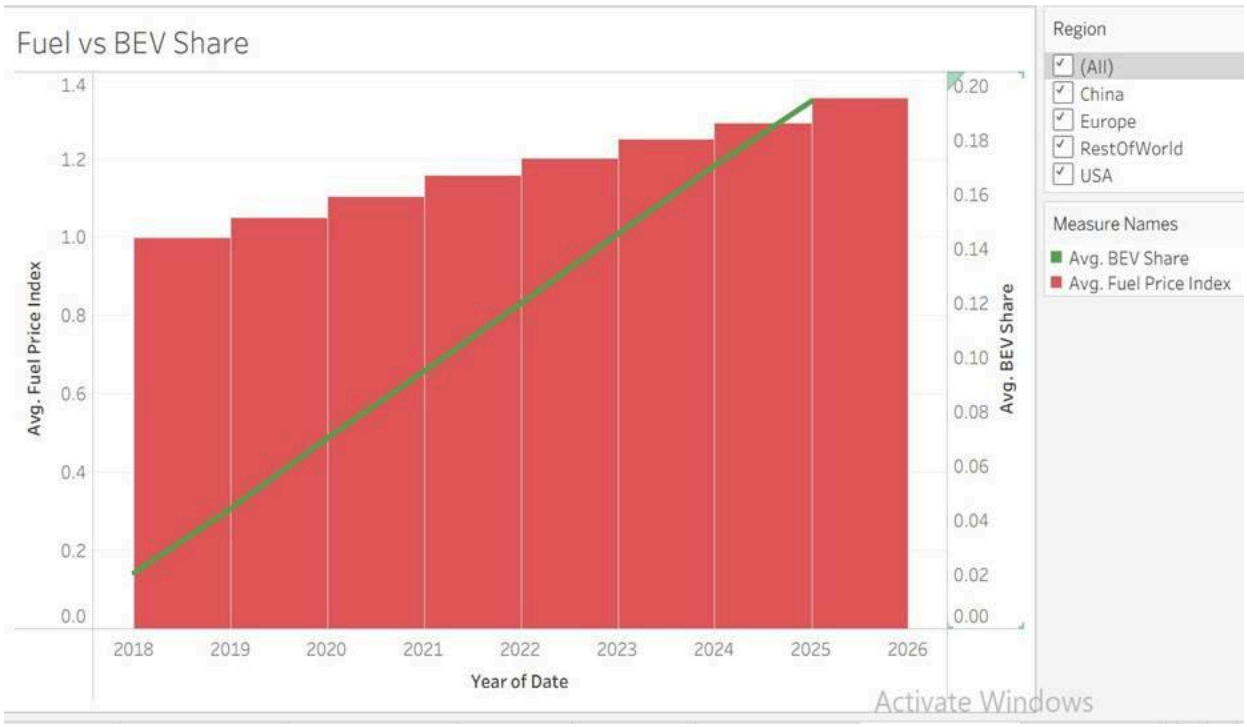

Figure 7: *Fuel Price Index vs BEV Share — Dual-Axis Chart (2018–2025)*

This chart plots the average fuel price index (red bars) against the average BEV share (green line) from 2018 to 2025. Both measures increase steadily throughout the period, and their parallel trajectories suggest a positive relationship between rising fuel costs and accelerating BEV adoption. As fuel prices rose from an index of 1.0 in 2018 to approximately 1.35 by 2025, BEV share grew from near 0% to approximately 20%, indicating that consumers responded to higher running costs of combustion vehicles by shifting toward electric alternatives.

4.8 Average Price Trend per Model

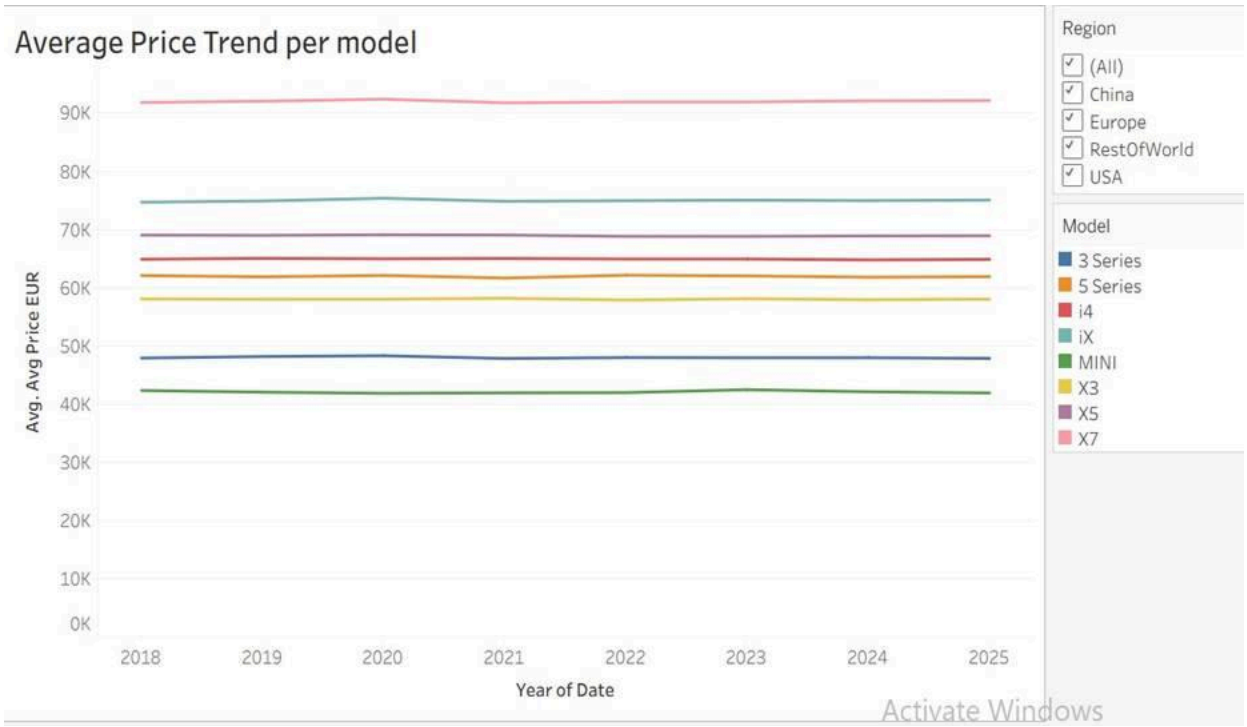

Figure 8: *Average Price Trend per Model — Multi-Line Chart (2018–2025)*

The chart tracks average transaction prices for all eight BMW models from 2018 to 2025, with each coloured line representing a different model. The X7 (pink) commands the highest average price at approximately €92K throughout the period, while MINI (green) sits at the lowest at around €42K, clearly illustrating BMW Group's broad price positioning strategy. Prices across all models remain remarkably stable over the eight-year period, with only minor fluctuations, suggesting BMW maintained disciplined pricing rather than discounting volume.

4.9 Units Sold Forecast (2026–2027)

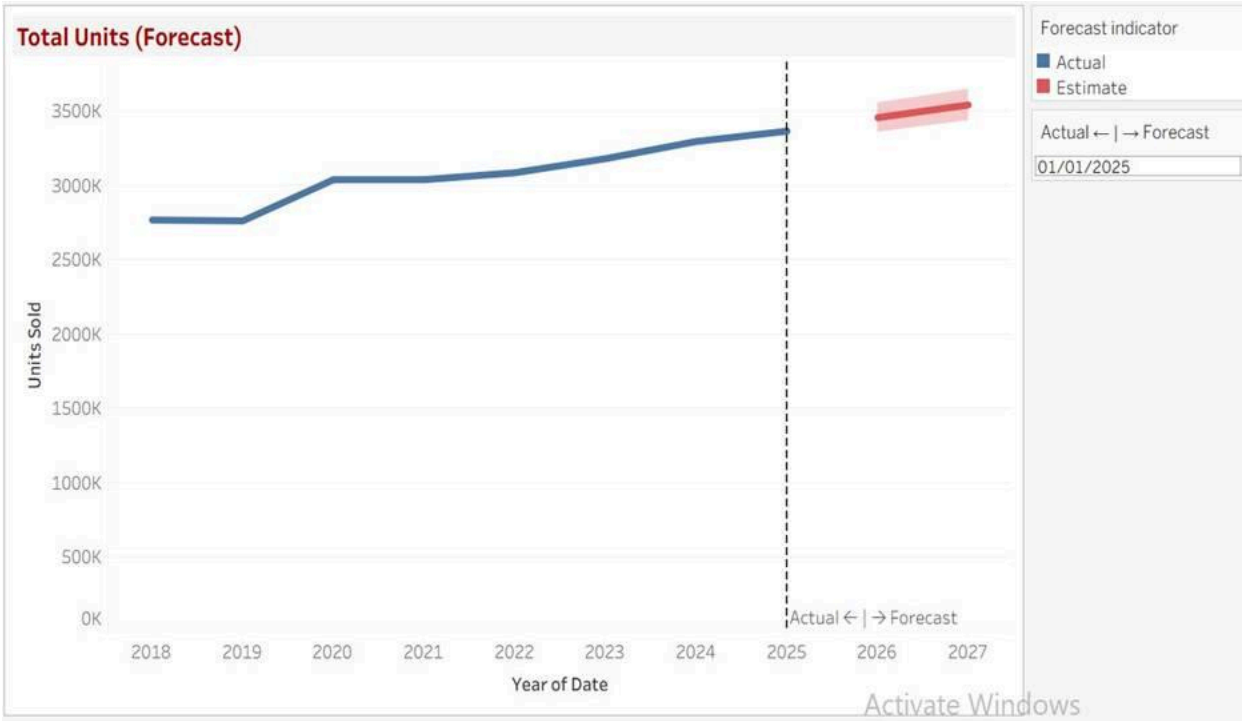

Figure 9: *Total Units Sold — Actual (2018–2025) and Forecast (2026–2027)*

The forecast chart extends BMW's total unit sales trajectory beyond the dataset using Tableau's built-in exponential smoothing model. Actual sales (blue line) grew from approximately 2.8M in 2018 to 3.3M in 2025, representing consistent positive growth. The forecast (red line with shaded confidence interval) projects a modest continuation of growth into 2026, with the confidence band widening toward 2027 to reflect increasing projection uncertainty.

4.10 Model Rankings Over Time

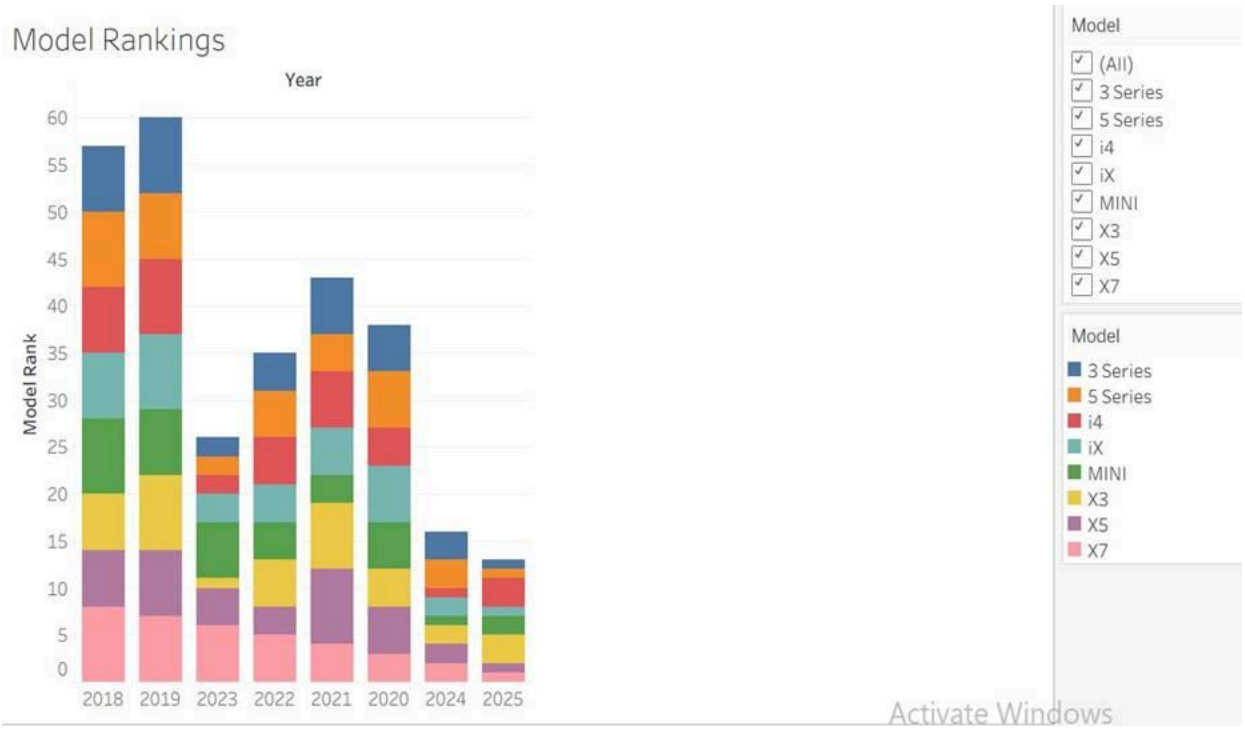

Figure 10: *Model Rankings — Stacked Rank Chart (2018–2025)*

This chart shows how BMW's model sales rankings changed year by year from 2018 to 2025. The 3 Series (dark blue) sits at the bottom of every bar throughout the entire period, indicating it consistently held the top sales ranking across all eight years — suggesting it is BMW's most reliably popular model globally. The decreasing bar heights from 2018–2019 toward 2024–2025 suggest that the gap between models is getting smaller over time, meaning other models are gradually catching up.

4. Conclusion

This exploratory data analysis of BMW Group's global sales data from 2018 to 2025 reveals a business in consistent growth across all four regions, with China emerging as the largest and fastest growing market.

On BEV adoption, China leads all regions by a significant margin, with all regions showing exponential growth suggesting the EV transition is accelerating. The fuel price analysis further suggests that rising fuel costs are contributing to faster EV switching behaviour among consumers.

On GDP and sales, BMW proved surprisingly resilient to mild economic slowdowns — most notably in 2025 where sales hit their highest point ever despite GDP growth being at its lowest. The only significant sales drop was in 2020, caused by COVID-19 lockdown disruptions rather than economic conditions alone.

On pricing, the X7 consistently commands the highest average price while the 3 Series maintained the top sales ranking throughout the entire period. BMW has not relied on discounting to drive growth, with prices remaining stable across all models from 2018 to 2025.

On seasonality, March and December are consistently the highest revenue months across most regions, while the USA shows a notably more stable year-round buying pattern compared to China and Europe.

Taken together, the eleven visualizations confirm that BMW is a resilient and growing premium business, with China as its most important market and electric vehicle adoption as its defining strategic challenge going forward.

5. Appendix — Dataset Overview

The dataset used in this analysis is bmw_global_sales_2018_2025.csv, containing monthly sales records for BMW Group across four global regions and eight vehicle models spanning January 2018 to December 2025.

Key variables included in the dataset:

- Year and Month — time dimension for all temporal visualizations
- Region — categorical: China, Europe, USA, RestOfWorld
- Model — categorical: 3 Series, 5 Series, X3, X5, X7, iX, i4, MINI
- Units Sold — primary sales volume metric
- Avg Price EUR — average transaction price per model per region
- Revenue EUR — estimated revenue (Units Sold × Avg Price)
- BEV Share — proportion of battery electric vehicles in total sales
- Premium Share — proportion of premium-tier vehicles
- GDP Growth — regional macroeconomic growth indicator
- Fuel Price Index — indexed fuel price relative to 2018 baseline

All visualizations were created using Tableau Public. No external data sources were used beyond the provided dataset.